

Supporting Information

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Fast Nucleation and Growth of ZIF-8 Nanocrystals Monitored by Time-Resolved In Situ Small-Angle and Wide-Angle X-Ray Scattering**

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1. Chemicals and Synthesis of ZIF-8 Nanocrystals

Commercially available reagents were used as received: $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (Sigma-Aldrich, $\geq 99.0\%$), 2-methylimidazole (Sigma-Aldrich, 99.0%), methanol (Sigma-Aldrich, $\geq 99.8\%$).

For the in situ SAXS/WAXS experiments the recently reported protocol of ZIF-8 nanocrystal synthesis at room temperature^[1] was adopted: Typically, 734.4 mg (2.469 mmol) of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and 810.6 mg (9.874 mmol) of 2-methylimidazole (Hmim) are each dissolved in 50 mL of MeOH. The latter clear solution is poured into the former clear solution under stirring with a magnetic bar. Stirring is stopped after combining the two solutions. The mixture becomes turbid within a few minutes. The nanocrystals are separated after 1 h (or 24 h) by centrifugation and washed with fresh MeOH. The centrifugation and washing step is repeated three times. A detailed characterization of the nanocrystals is provided elsewhere.^[1]

- [1] a) J. Cravillon, S. Münzer, S.-J. Lohmeier, A. Feldhoff, K. Huber, M. Wiebcke, *Chem. Mater.* **2009**, *21*, 1410; b) J. Cravillon, R. Nayuk, S. Springer, A. Feldhoff, K. Huber, M. Wiebcke, *Chem. Mater.* **2011**, DOI: 10.1021/cm103571y.

2.1. Time-Resolved In Situ SAXS/WAXS Experiments

Time-resolved in situ SAXS/WAXS experiments were performed at beamline ID02 of the European Synchrotron Radiation Facility (ESRF, Grenoble, France). The wavelength was set at $\lambda = 0.0995$ nm. Two-dimensional SAXS and WAXS patterns were recorded simultaneously on CCD-based detectors. In order to enable the detection of the smallest particles the SAXS detector was set at a distance close to the sample (1.077 m). For SAXS and WAXS the acquisition setup was synchronized with a Biologic SFM-400 stopped-flow device.^[2] The device was used for rapid turbulent mixing of the two component solutions (see Section 1.) through a mixing chamber followed by injection with a rate of $2 \text{ mL} \cdot \text{s}^{-1}$ into a quartz glass capillary of 1.5 mm diameter and immobilization by a hard-stop system. The glass capillary served as the scattering cell. An external circulating fluid bath was used to keep the temperature at 25°C . The stopped-flow device was equipped with four motorized syringes with controllable pumping speed containing the two component solutions as well as 1 M HCl acid and distilled water. The acid and water were used for cleaning the mixer and scattering cell after each experiment.

- [2] P. Panine, S. Finet, T. M. Weiss, T. Narayanan, *Adv. Colloid Interface Sci.* **2006**, *127*, 9.

2.2. Evaluation of SAXS Data

Measured 2D time-resolved SAXS patterns were corrected for dark level, sensitivity of the CCD detector and normalized to an absolute scale considering the intensity of the transmitted beam, the scattering from a secondary water standard and the thickness of the capillary. The normalized 2D SAXS patterns were azimuthally averaged to obtain the 1D scattering profiles, denoted as differential scattering cross section ($d\Sigma/d\Omega$) in units of reciprocal millimeters versus momentum transfer q

$$q = \left(\frac{4\pi}{\lambda}\right) \sin\theta \quad (1)$$

with λ being the wavelength of the X-rays and 2θ being the scattering angle under which $(d\Sigma/d\Omega)$ was observed. Appropriate background subtraction of the solvent (MeOH)

$$I = \left(\frac{d\Sigma}{d\Omega}\right)_{\text{sample}} - \left(\frac{d\Sigma}{d\Omega}\right)_{\text{solvent}} \quad (2)$$

resulted in the net differential scattering cross section $I(q)$ of the solute.

The SAXS patterns beyond 60 s could not be adequately evaluated, because the diameter of the ZIF-8 particles after 60 s comes close to the resolution limit $d_{\text{max}} = 31$ nm of our SAXS setup ($d_{\text{max}} < \pi \cdot q_{\text{min}}^{-1}$)^[3]. The scattering patterns before 60 s do not show a maximum as is typically observed for systems with strong particle-particle correlations. This is in agreement with the rather low values of the number density and volume fraction of the particles at 60 s (see Section 2.3 for a quantitative estimation of these parameters). Therefore, we could neglect inter-particle interactions (i.e. the structure factor, $S(q)$, was set to $S(q) = 1$).

- [3] L. A. Feigin, D. I. Svergun, *Structure Analysis by Small-Angle X-Ray and Neutron Scattering*, Plenum Press, New York, **1987**, pp. 82-83.

2.2.1 Model-Independent SAXS Data Evaluation

For a typical scattering curve of particles a characteristic decay of the signal to zero values is expected with increasing q which would eventually lead to noisy data points provided that the q regime is large enough. However, in the case of the ZIF-8 system there is a signal beyond $q = 1 \text{ nm}^{-1}$, which does not decay below $5 \cdot 10^{-4} \text{ nm}^{-1}$. This signal is attributed to clusters and/or smaller building units and ions. We approximate the intercept of this signal at $q = 0$ by using an average signal value established at the q range of $0.980 \text{ nm}^{-1} \leq q \leq 1.023 \text{ nm}^{-1}$ of the corresponding scattering curve. Justification of this value is based on the assumption that the scattering contribution from these smaller units has already reached a plateau value at $q \leq 1 \text{ nm}^{-1}$. Indeed, scattering curves recorded before 15 s show only scattering contributions from the clusters and smaller building units and reach a plateau value below $q \approx 1 \text{ nm}^{-1}$ (see the scattering curve measured at 12 s shown in Figure S5), which confirms the validity of this assumption. In order to treat the growing ZIF-8 particles, this residual signal stemming from the clusters and/or from smaller building units and ions was subtracted (see Figure S1).

Scattering data after subtraction of the contribution from the clusters and smaller building units and ions were used for a model-independent interpretation.^[4] This model-independent analysis includes a Guinier evaluation and the calculation of the Porod invariant.

The initial parts of the experimental curves were processed using the Guinier approximation according to

$$\ln(I) \propto \ln(I(q = 0)) - \left(\frac{R_g^2}{3}\right) \cdot q^2 \quad (3)$$

I is the absolute scattering cross section as a function of the momentum transfer q . $I(q = 0)$ is the absolute scattering cross section at $q = 0$. The latter value is proportional to the weight-

averaged molar mass of the ZIF-8 particles excluding clusters, building units and ions. R_g^2 is the z -averaged squared radius of gyration of the ZIF-8 particles. Figure S2 provides two

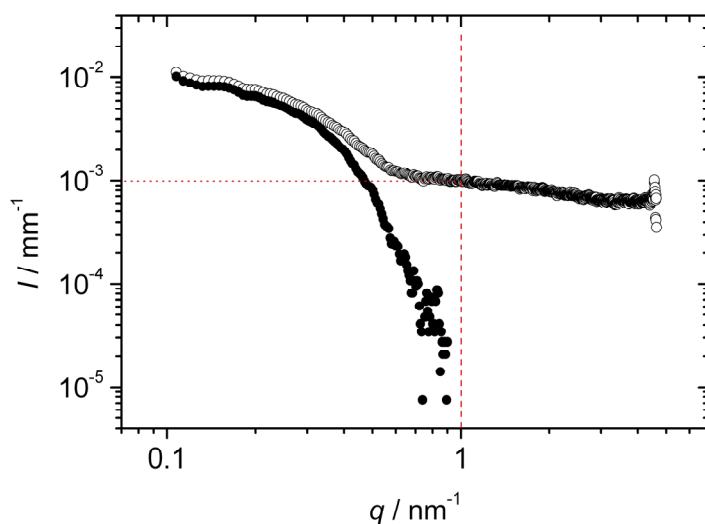


Figure S1. Scattering curve at 25 s before (hollow circles) and after (filled circles) subtraction of the signal from the clusters and/or building units and ions. The horizontal dotted line shows the contribution of the clusters and building units to the total scattering of the sample to be subtracted. As indicated by the vertical dashed line, experimental data points for $q \geq 1 \text{ nm}^{-1}$ were cut off after subtraction.

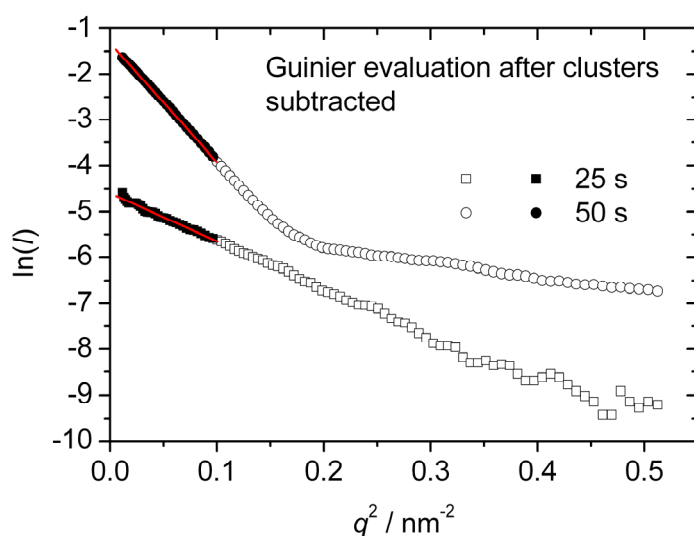


Figure S2. Examples of Guinier analysis of the scattering curves measured at 25 s (circles) and 50 s (squares) after subtraction of the contribution from the clusters, building units and ions. Sections of the curves indicated with the filled symbols were used for the Guinier approximation (red lines).

examples of such a Guinier analysis. The q -range used for the Guinier evaluation was fixed and covered the regime $0.1075 \text{ nm}^{-1} \leq q \leq 0.3103 \text{ nm}^{-1}$.

The Porod invariant Q is proportional to the mass concentration of all scattering species in general. It corresponds to the area under the curve $4\pi I q^2$ versus q (Kratky plot) with $q \rightarrow \infty$ (see Figure S3)

$$Q = \int_q 4 \cdot \pi \cdot I \cdot q^2 dq. \quad (4)$$

Since we are interested in the contribution from the ZIF-8 particles only, the respective invariant Q is established from the scattering curves after having subtracted the scattering contribution from the clusters, building units and ions. As indicated on Figure S3, the missing initial part of the curve was approximated by a triangle. The invariant of the particles enables us to extract the weight-averaged mass values of the particles only and the number density of particles. The evaluation of these two parameters is as follows: As outlined before, $I(q=0)$ as the intercept of a scattering curve is proportional to the weight average molar mass of the ZIF-8 particles. In addition, $I(q=0)$ is proportional to the mass concentration of the particles. After dividing this $I(q=0)$ by Q

$$M_p \propto \frac{I(q=0)}{Q} \quad (5)$$

we obtain the value proportional to a weight average molecular weight of the particles only (M_p). When dividing the Porod invariant Q (proportional to the mass concentration of the particles) by the right hand side of eq. (5) (proportional to M_p of the particles) we obtain the value proportional to the number density (N/V) of all ZIF-8 particles N

$$\frac{N}{V} \propto \frac{Q}{M_p} = \frac{Q^2}{I(q=0)} \quad (6)$$

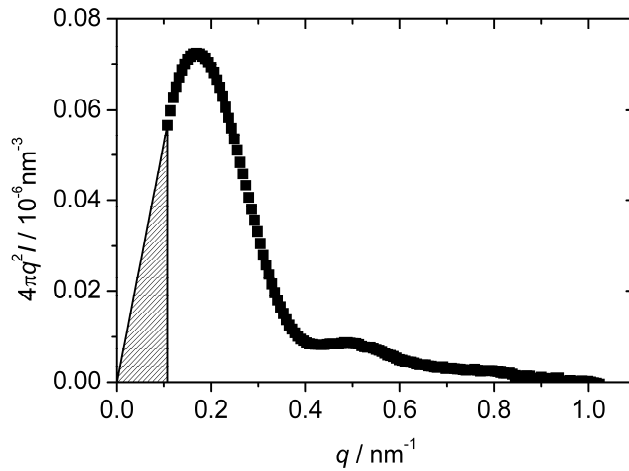


Figure S3. Kratky plot of the scattering curve at 50 s after subtraction of the contribution from the clusters, building units and ions.

It has to be emphasized at this point, that an analogous data reduction can be performed with the scattering curves prior to subtracting the scattering contribution from the clusters. In the example represented in Figure S1, the fit would thus be applied to the data indicated by hollow circles. These alternative fits lead to values for a z-averaged squared radius of gyration R_g^2 and a weight averaged mass M_w , which average over all solid species including the ZIF-8 particles, clusters and smaller units. A correlation of the respective radii versus the mass values is shown in Figure S4. The correlation results in an exponent of $\beta = 0.17$ for the power law $R_g \sim M_w^\beta$. This value dropped by a factor of two if compared to the value which is discussed in the manuscript and which is based on scattering data after subtraction of the cluster contribution and successive division by the invariant Q . As has been outlined elsewhere,^[4] such a drop by a factor of two is observed for a growth process which follows a monomer addition mechanism but not for a coalescence process.

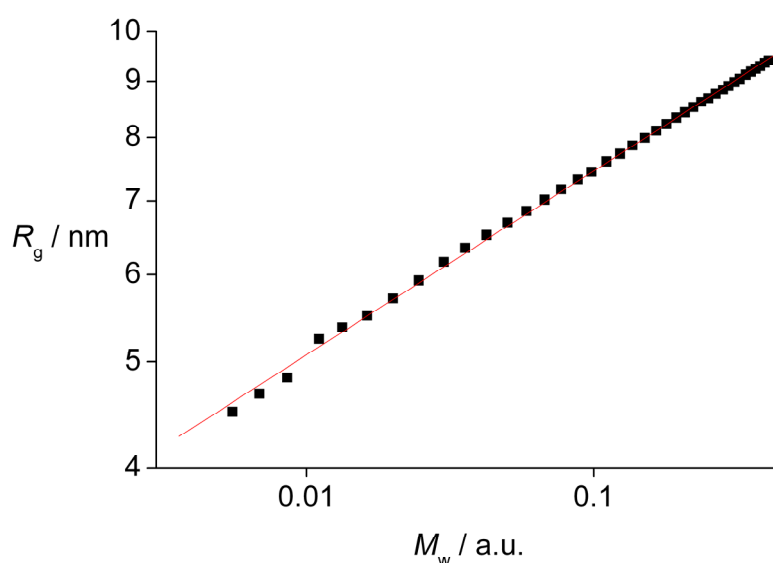


Figure S4. Correlation of the radius of gyration (R_g) and the weight-averaged molar mass (M_w) revealing power law behavior. The slope of the straight line gives an exponent $\beta = 0.17$.

- [4] J. Liu, S. Pancera, V. Boyko, A. Shukla, T. Narayanan, K. Huber, *Langmuir* **2010**, 26, 17405.

2.2.2 Model-Dependent SAXS Data Evaluation

Model fitting of background-corrected scattering patterns was performed using the IGOR Pro-based SANS analysis software (version 6.011) provided by the NIST Center for Neutron Research (NCNR).^[5]

The contributions from the clusters and particles in the scattering curves at various times were modelled as follows:

Up to 14 s, the scattering curves show only an intensity $I(q)$ originating from clusters and smaller building units and ions. A model of monodisperse homogeneous spheres was used.

Between 15 and 30 s, the scattering curves show an intensity originating from clusters and particles. A bimodal model consisting of monodisperse spheres (for the clusters) and polydisperse spheres with a Schulz size distribution (for the particles) was used.

Between 31 and 60 s, the scattering curves again show intensity originating from particles and clusters and building units and ions, but the intensity originating from the clusters and building units and ions has decreased and it was found that the model fits could be improved if a bimodal model was used consisting of polydisperse spheres with a Schulz size distribution as above (for the particles) and an additive Lorentzian function for random fluctuations (for the clusters and/or building units and ions).

Figure S4 shows selected scattering curves with model fits.

The differential scattering cross section of monodisperse spheres $I(q,R)$ can be expressed as

$$I(q,R) = scale \cdot P(q,R) = scale \cdot \left(\frac{3 \cdot (\sin(q \cdot R) - q \cdot R \cdot \cos(q \cdot R))}{(q \cdot R)^3} \right)^2 \quad (7)$$

with $P(q,R)$ being the form factor of a sphere with the radius R .

The Schulz size distribution $f(R)$ expressed as molar fraction of the particles with size R is given by^[6]

$$f(R) = (z+1)^{z+1} \cdot \left(\frac{R}{R_{av}} \right)^z \cdot \frac{e^{-\frac{R}{R_{av}} \cdot (z+1)}}{R_{av} \cdot \Gamma(z+1)} \quad (8)$$

with R_{av} being the average radius. The polydispersity p is related to z and the variance σ^2 of the distribution by

$$p = \sigma/R_{av} \text{ and } z = \left(\frac{1}{p^2} \right) - 1. \quad (9)$$

The scattering intensity $I(q,R)$ for polydisperse spheres becomes

$$I(q,R) = scale \cdot \int_0^\infty f(R) \cdot R^6 \cdot P(q,R) dR. \quad (10)$$

An analytical solution of the integral can be found elsewhere.^[7]

The radius of gyration R_g can be calculated by

$$R_g = \sqrt{\frac{3 \langle R^8 \rangle}{5 \langle R^6 \rangle}} = R_{av} \sqrt{\frac{3 \cdot (z+8) \cdot (z+7)}{5 \cdot (z+1)^2}} \quad (11)$$

with $\langle R^N \rangle$ being the N^{th} moment of the size distribution

$$\langle R^N \rangle = \frac{R_{av}^N}{(z+1)^N} \cdot \frac{(z+N)!}{z!}. \quad (12)$$

The additive Lorentzian function is given by^[8]

$$S(q) = \frac{S(0)}{1+(q\xi)^2} \quad (13)$$

with ξ being a correlation length that is smaller than the spherical particles.

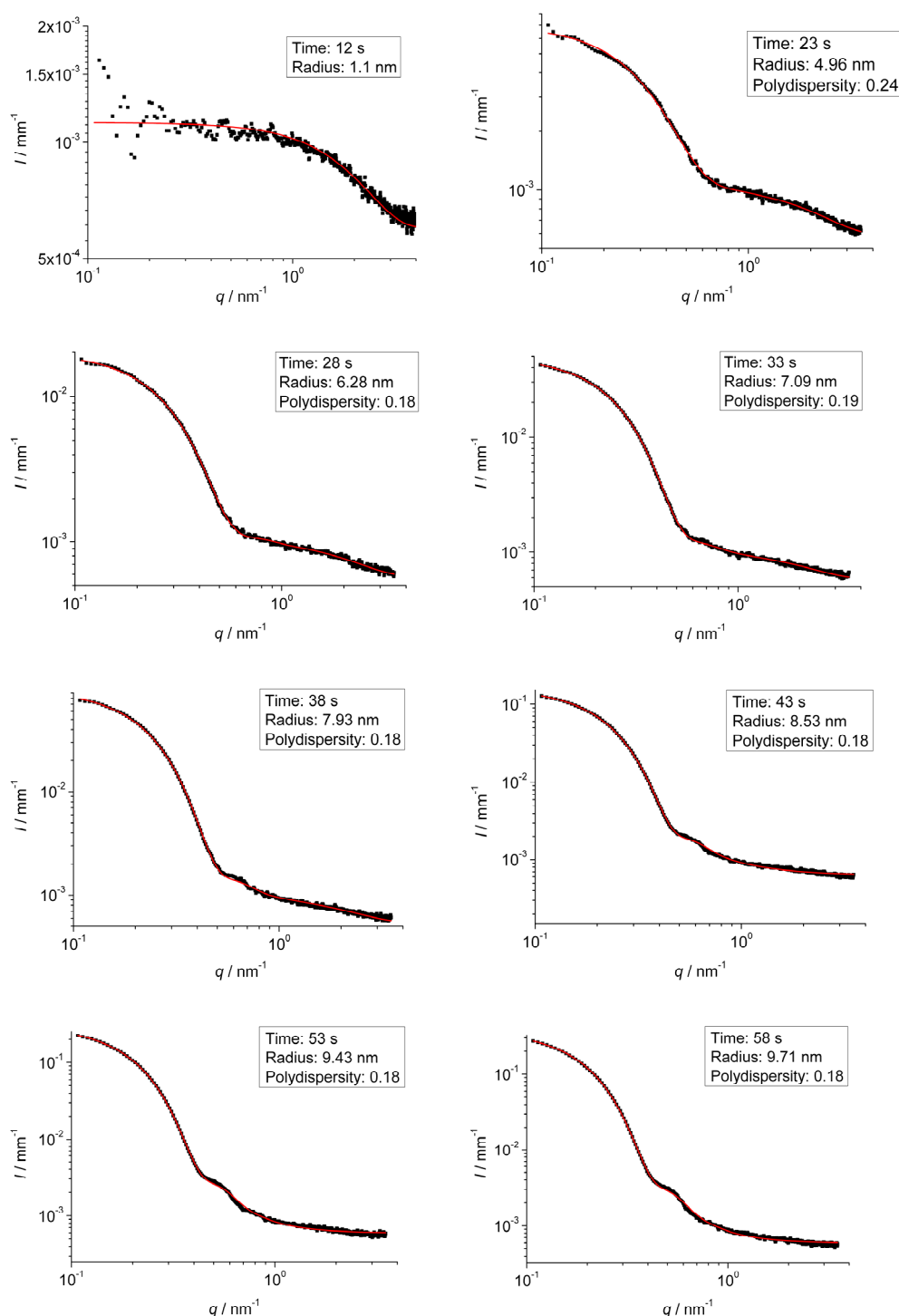


Figure S5. Selected experimental SAXS patterns (black squares) with model fits (red lines). In the pattern at 12 s the radius of the clusters (monodisperse spheres) is given, whereas in the patterns at times larger than 14 s the respective average radius R_{av} of the particles (polydisperse spheres) is given.

- [5] S. R. J. Kline, *J. Appl. Cryst.* **2006**, 39, 895.
- [6] G. V. Schulz, *Z. Phys. Chem.* **1935**, 43, 25.
- [7] M. Kotlarchyk, S.-H. Chen, *J. Chem. Phys.* **1983**, 79, 2461.
- [8] L. S. Ornstein, F. Zernike, *Proc. Sect. Sci. K. Med. Akad. Wet.* **1914**, 17, 793.

2.3 Results of SAXS Data Evaluation

Figure S6 shows the average radius (R_{av}) and polydispersity (p) of the ZIF-8 particles as obtained by model fitting as a function of time. The polydispersity remains nearly constant between 22 and 60 s. In Figure S7, the radii of gyration (R_g) as obtained from model-independent and model-dependent data analysis, respectively, are compared. The values are in close agreement with each other.

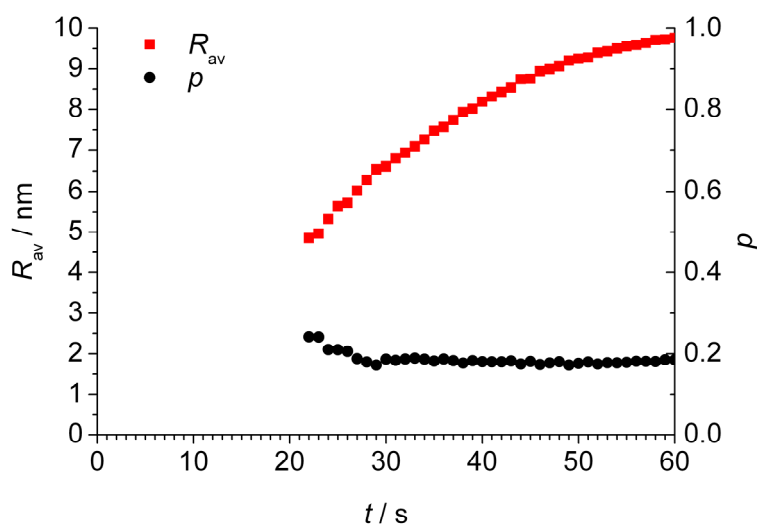


Figure S6. Average radius (R_{av}) and polydispersity (p) of the ZIF-8 particles as obtained from model fitting versus time (t).

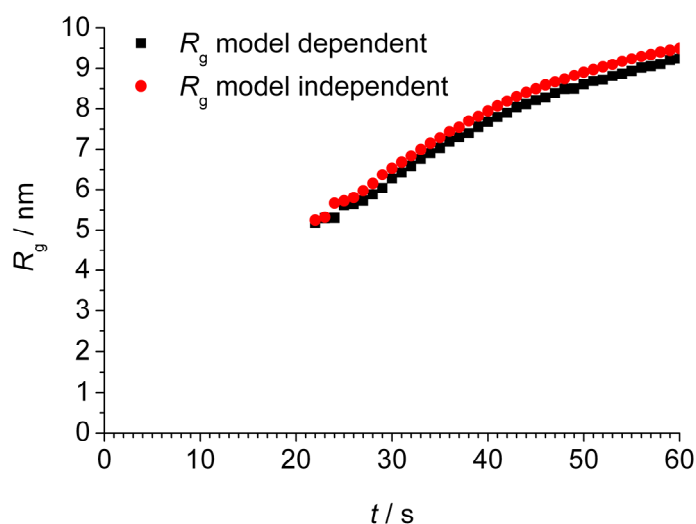


Figure S7. Radii of gyration (R_g) of the ZIF-8 particles as obtained from model-dependent and model-independent SAXS data analysis versus time (t).

A more quantitative interpretation of the present data becomes possible if we assume for the ZIF-8 nanocrystals a composition of $[\text{Zn}(\text{mim})_2] \cdot 2\text{MeOH}$ and a cubic unit cell volume of $V_{\text{EZ}} = 4.923 \text{ nm}^3$ (density $\rho_{\text{ZIF-8}} = 1.193 \text{ g} \cdot \text{cm}^{-3}$)^[9]. This yields a scattering lengths density for the nanocrystals of $\sigma_{\text{ZIF-8}} = 10.587 \cdot 10^{-4} \text{ nm}^{-2}$ and a scattering lengths density contrast against the solvent MeOH ($\sigma_{\text{MeOH}} = 7.536 \cdot 10^{-4} \text{ nm}^{-2}$) of $\Delta\sigma = 3.051 \cdot 10^{-4} \text{ nm}^{-2}$. Using this value, the number density (N/V) and volume fraction (Φ) of the nanocrystals can be calculated according to

$$I(q = 0) = N/V \cdot \langle V_p^2 \rangle \cdot (\Delta\sigma)^2 = \Phi \cdot \langle V_p \rangle \cdot (\Delta\sigma)^2 \quad (14)$$

with $\langle V_p \rangle$ being the average volume of the particles. At a time of 60 s, the calculation yields $(N/V) = 1.14 \cdot 10^{11} \text{ mm}^{-3}$ and $\Phi = 4.4 \cdot 10^{-4}$. The maximum volume fraction (Φ_{max}) may be calculated by $\Phi_{\text{max}} = C_0 \cdot V_m$ with C_0 being the initial concentration of Zn^{2+} cations in solution and V_m the molar volume of ZIF-8, yielding $\Phi_{\text{max}} = 6.1 \cdot 10^{-3}$. This allows to calculate the yield (with respect to Zn^{+2}) of the reaction at any time by $Y = 100 \cdot \Phi / \Phi_{\text{max}}$. At 60 s, a yield of 7.2 % is estimated.

- [9] X.-C. Huang, Y.-Y. Lin, J.-P. Zhang, X.-M. Chen, *Angew. Chem.* **2006**, *118*, 1587; *Angew. Chem. Int. Ed.* **2006**, *45*, 1557.

3.1. Time-Resolved In Situ Static Light Scattering (SLS) Experiments

SLS was performed with a home built multi-angle goniometer described by Becker and Schmidt.^[10] Cylindrical silica glass cuvettes with a diameter of 20 mm from Hellma (Mülheim, Germany) served as scattering cells. This goniometer was equipped with a He-Ne laser operating at a wavelength of 632.8 nm and allowed simultaneous recording of the scattering intensity at 2 times 19 scattering angles. The angular regime covered a range of $25.84^\circ \leq \theta \leq 143.13^\circ$. Recording of an angular dependent curve was completed after 2 ms. 1000 successive recordings were added to form one measurement requiring 8 s in total. The time interval between the start of two successive measurements was 10 s. The measuring temperature was 25 °C.

All experiments were based on two component solutions (see Section 1.): (i) 24.6 mmol $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ dissolved in 1 L of methanol and (ii) 98.7 mmol of Hmim dissolved in 1 L of MeOH. The solutions were cleaned by passing them through 0.46 μm filters to remove any dust particles. The filtered component solutions were injected into the scattering cuvette by applying a SFM-200 Biologic stopped-flow device. Homogeneous mixing was performed at a flow-rate of the component solutions of $2 \text{ mL} \cdot \text{s}^{-1}$, in order to reproduce the mixing conditions used for the SAXS/WAXS experiments (same injection rate). Filling of the cuvette took ~ 5 s.

- [10] A. Becker, M. Schmidt *Macromol. Chem. Macromol. Symp.* **1991**, *50*, 249.

3.2. Evaluation of SLS Data

Scattering curves were processed as the Rayleigh ratio ΔR_θ at variable scattering angle θ . The Rayleigh ratio of the particles were calculated from the difference between the Rayleigh ratio of the particle dispersion and of the solvent. Details are given elsewhere.^[11] Scattering curves were evaluated according to the Guinier approximation (see Figure S8)

$$\ln\left(\frac{K_c}{\Delta R_\theta}\right) \propto \ln\left(\frac{1}{M_w}\right) + \frac{R_g^2}{3} \cdot q^2 \quad (15)$$

In eq. (15) K_c is the contrast factor of the scattering species forming in solution. These species are assumed to consist of ZIF-8. ΔR_θ is the Rayleigh scattering of the ZIF-8 particles at the scattering angle θ , c is the concentration of ZIF-8 building units given by the formulae $[\text{Zn}(\text{mim})_2]$ in $\text{g}\cdot\text{L}^{-1}$. The concentration of the building units was calculated with respect to Zn^{2+} . M_w is the weight average molar mass, and R_g^2 is the z -averaged squared radius of gyration of the growing particles. The contrast factor

$$K = \frac{4 \cdot \pi^2 \cdot n^2}{N_A \cdot \lambda^4} \cdot \left(\frac{dn}{dc}\right)^2 \quad (16)$$

and the momentum transfer

$$q = \frac{4 \cdot \pi \cdot n}{\lambda} \cdot \sin\left(\frac{\theta}{2}\right) \quad (17)$$

were calculated with N_A Avogadro's number, with $\lambda = 632.8$ nm the laser wavelength, with $n = 1.326$ the refractive index of pure methanol as solvent and with dn/dc the refractive index increment of ZIF-8 in methanol. Lack of the dn/dc forced us to apply a default value of $dn/dc = 0.1 \text{ mL}\cdot\text{g}^{-1}$. This still reproduces the correct order of magnitude of mass values and gives correct relative values.

In addition, size and mass of ZIF-8 particles depends in a complex way on the growth time. Therefore, an appropriate extrapolation of $\frac{K_c}{\Delta R_\theta}$ to $c = 0$ was not possible. However, neglect of the concentration dependent term by using eq. (15) seems to be justified in light of the low concentrations applied in the present work.

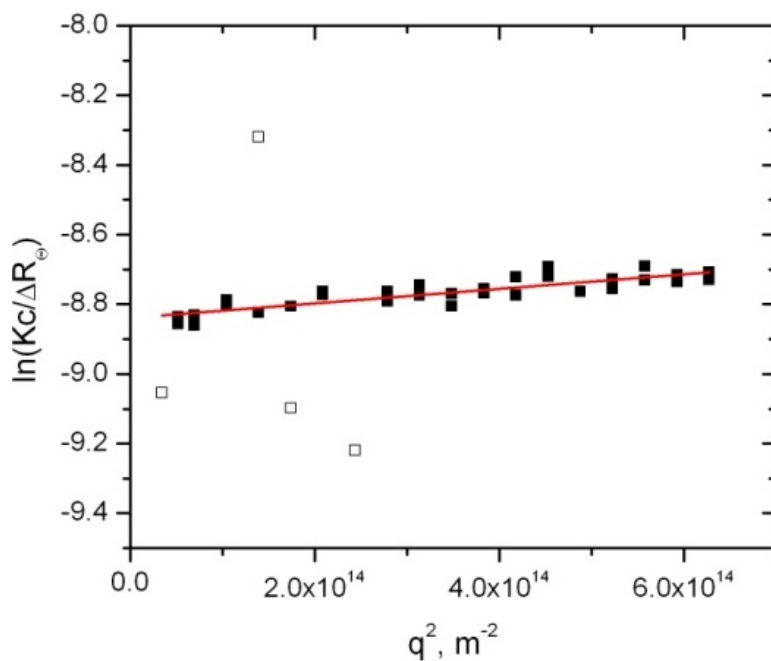


Figure S8. Representative example of a scattering curve evaluated according to eq. (15).

[11] J. Liu, J. Rieger, K. Huber *Langmuir* **2008**, 24, 8262.

3.3. Results of SLS Data Evaluation

Figure S9 and Figure S10 show the weight-averaged molar mass and the radius of gyration as a function of time for a typical SLS measurement.

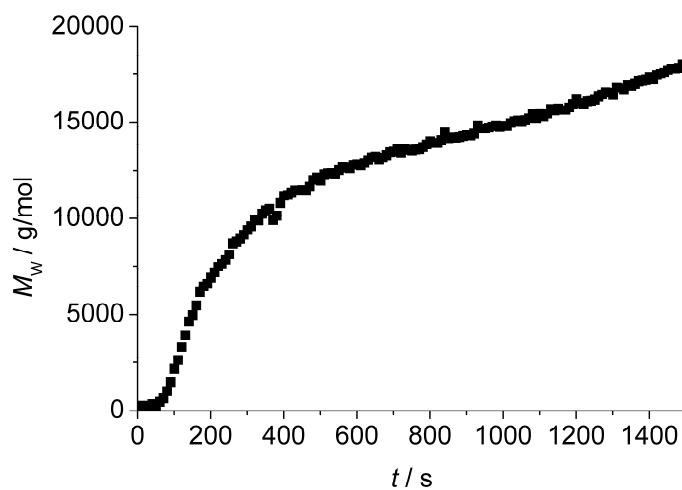


Figure S9. Weight-averaged molar mass (M_w) versus time (t).

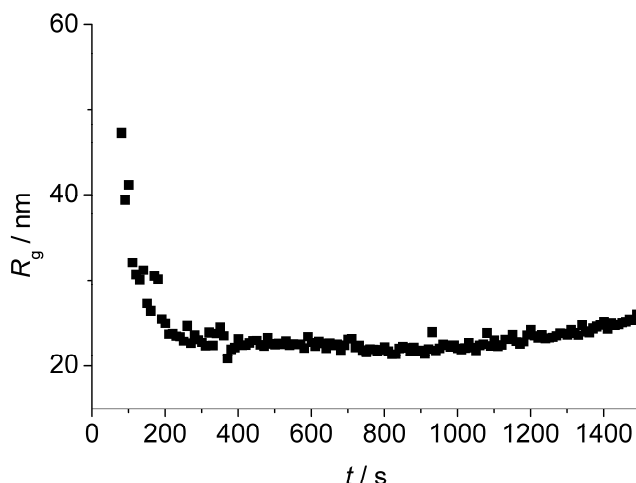


Figure S10. Radius of gyration (R_g) versus time (t).

The first significant detection of a weight-averaged particle mass (M_w) and particle size (R_g) becomes possible by SLS after ~ 100 s and ~ 150 s, respectively. Thus, the very early stages cannot be monitored by SLS. While the particle size remains constant at $R_g \approx 25$ nm (corresponding to $R_{\text{sphere}} = 33$ nm) up to ~ 1000 s, the mass values continuously increase with time. The increase of mass values becomes clearly slower at ~ 400 s. This situation of constant size but increasing mass values can be interpreted as follows: Single particles grow very fast compared to the time resolution of our SLS experiment and the individual growth becomes slow when a size close to $R_g \approx 25$ nm is attained. The increase in particle mass is predominantly due to a continuous and comparatively slow nucleation, which increases the number of particles with time. Beyond ~ 1000 s, both the particle mass and size values increase. This can be interpreted by a loose agglomeration of the primary nanoparticles which essentially keep their size. The size of the primary nanoparticles estimated by SLS (diameter ~ 66 nm) nicely agrees with the size determined by SEM (diameter ~ 55 nm, Figure 2). The same situation has been observed also by SLS when component solutions of the same compositions were mixed by hand as is done for nanocrystal preparation (see Section 1.)^[1], confirming that turbulent mixing and usual mixing conditions result in very similar crystallization reactions and therefore yield final primary nanocrystals of approximately the same size. As shown elsewhere,^[1] the primary nanocrystals slowly continue to grow for up to 24 h, most likely by Ostwald ripening.

These SLS observations are in qualitative agreement with the observations made by our SAXS/WAXS experiments using turbulent mixing. In both cases continuous nucleation and fast particle growth ran parallel for an extended period of time. However, the processes monitored in the SAXS/WAXS glass capillary under stopped-flow conditions deviate quantitatively from those observed in the much larger SLS cuvette as they are faster with a higher nucleation rate and, accordingly, a shorter nucleation period (up to ~ 60 s) leading to smaller primary nanocrystals (diameter ~ 25 nm).

4. SEM Experiments

Scanning electron micrographs were taken in secondary electron contrast at an acceleration voltage of 2 keV using a JEOL JSM-6700F field-emission instrument. Samples were dispersed on a carbon sample holder. For size estimation, 300 particles were considered on the SEM micrographs.